IETF Hackathon

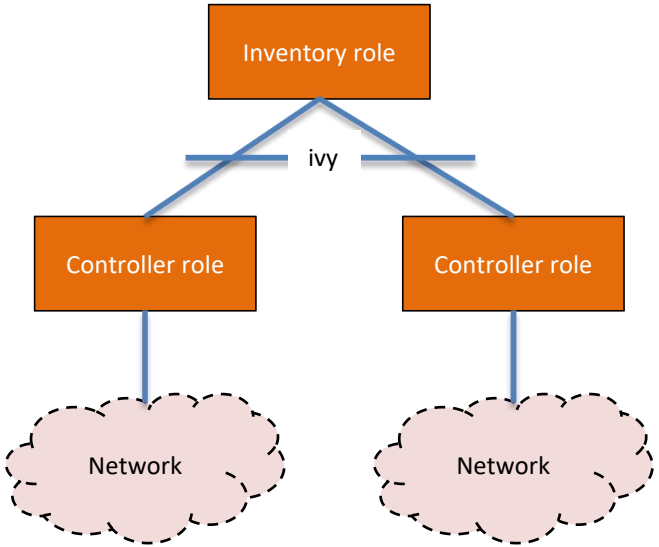
-YANG Data Model for Network Inventory (IVY)

IETF 125
14–15 March 2026
Shenzhen, China

I E T F

Hackathon Architecture

- Most likely configuration:
  - A controller from one vendor plays the role of inventory system (inventory role)
  - Several controllers from different vendors are plays the role of inventory controller (controller role)
  - Each controller role provides a view of network elements and their physical inventory
- If a vendor has an inventory system, this can take the inventory role



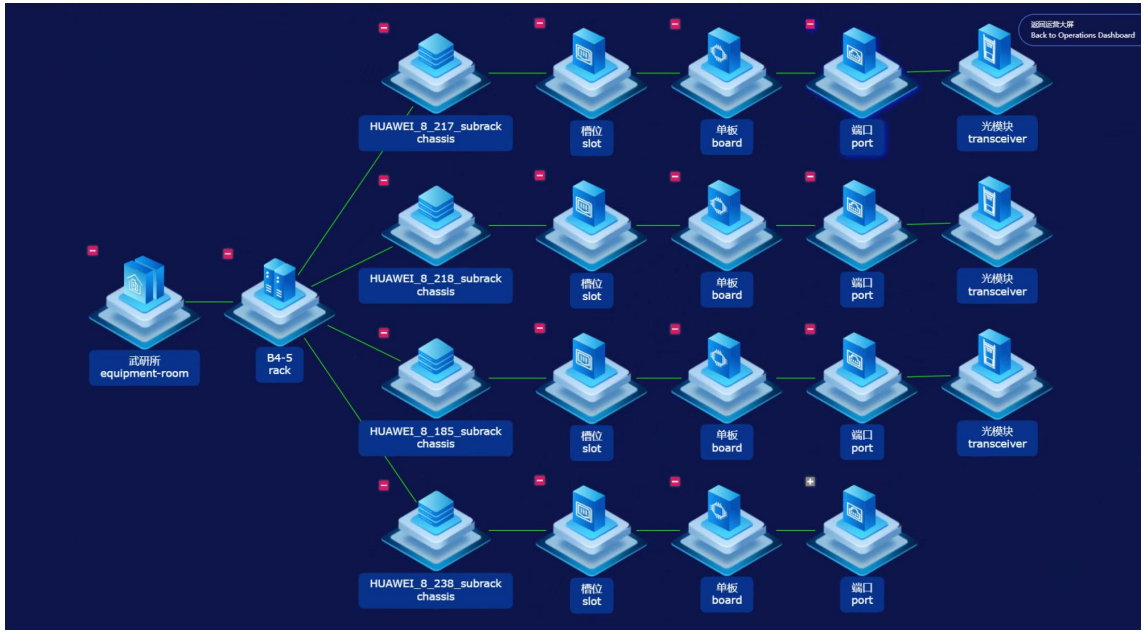
Network may be:
- An image in the controller
- A simulation
- Real hardware

Ideally network elements would be a mix of IP, Ethernet, OTN and optical where there were various NEs of different mixes of capabilities. This should have no impact on the ivy interface capability.

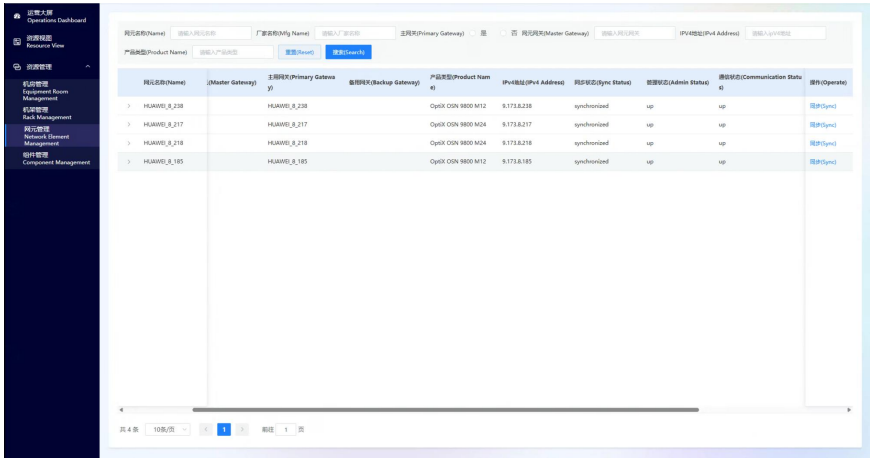
```
module: ietf-network-inventory
  +--ro network-inventory
    +--ro network-elements
      +--ro network-element* [ne-id]
        +--ro ne-id          string
        +--ro ne-type?       identityref
        +--ro uuid?          yang:uuid
        +--ro name?          string
        +--ro alias?         string
        +--ro description?   string
        +--ro software-rev* [name]
        |  +--ro name        string
        |  +--ro revision?   string
        |  +--ro patch* [revision]
        |     +--ro revision    string
        +--ro mfg-name?      string
        +--ro product-name?  string
        +--ro product-rev?   string
        +--ro components
          +--ro component* [component-id]
            +--ro component-id     string
            +--ro class            union
            +--ro uuid?            yang:uuid
            +--ro name?            string
            +--ro alias?           string
            +--ro description?     string
            +--ro software-rev* [name]
            |  +--ro name        string
            |  +--ro revision?   string
            |  +--ro patch* [revision]
            |     +--ro revision    string
            +--ro mfg-name?        string
            +--ro product-name?    string
            +--ro hardware-rev?    string
            +--ro mfg-date?        yang:date-and-time
            +--ro part-number?     string
            +--ro serial-number?   string
            +--ro asset-id?        string
            +--ro is-fru?          boolean
            +--ro uri*             inet:uri
            +--ro parent-rel-pos?  int32
            +--ro parent*
            |      -> ../../component/component-id
            +--ro is-main?         boolean
```

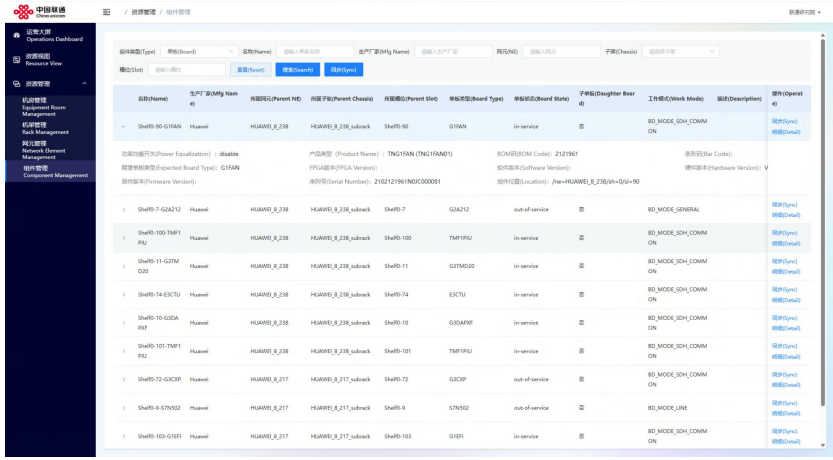
Ivy model

- <What draft were involved?>
  - A Base YANG Data Model for Network Inventory



Ivy system


运营大屏
Operations Dashboard
资源视图
Resource View
资源管理
机房管理
Equipment Room Management
机架管理
Rack Management
网元管理
Network Element Management
组件管理
Component Management
网元名称(Name)
厂家名称(Mfg Name)
主网元(Primary Gateway)
是
否
网元网关(Master Gateway)
IPV4地址(IPv4 Address)
产品类型(Product Name)
重置(Reset)
搜索(Search)
网元名称(Name)
(Master Gateway)
主用网关(Primary Gateway)
备用网关(Backup Gateway)
产品类型(Product Name)
IPv4地址(IPv4 Address)
同步状态(Sync Status)
管理状态(Admin Status)
通信状态(Communication Status)
操作(Operate)
HUAWEI_8_238
OptiX OSN 9800 M12
9.173.8.238
synchronized
up
HUAWEI_8_217
OptiX OSN 9800 M24
9.173.8.217
HUAWEI_8_218
9.173.8.218
HUAWEI_8_185
9.173.8.185
同步(Sync)
共 4 条
10条/页
1
前往
页


中国联通
China unicom
资源管理
组件管理
运营大屏
Operations Dashboard
资源视图
Resource View
资源管理
机房管理
Equipment Room Management
机架管理
Rack Management
网元管理
Network Element Management
组件管理
Component Management
组件类型(Type)
单板(Board)
名称(Name)
生产厂家(Mfg Name)
网元(NE)
子架(Chassis)
槽位(Slot)
重置(Reset)
搜索(Search)
同步(Sync)
名称(Name)
生产厂家(Mfg Name)
所属网元(Parent NE)
所属子架(Parent Chassis)
所属槽位(Parent Slot)
单板类型(Board Type)
单板状态(Board State)
子单板(Daughter Board)
工作模式(Work Mode)
描述(Description)
操作(Operate)
Shelf0-90-G1FAN
Huawei
HUAWEI_8_238
HUAWEI_8_238_subrack
Shelf0-90
G1FAN
in-service
否
BD_MODE_SDH_COMM ON
同步(Sync)
明细(Detail)
功率均衡开关(Power Equalization) : disable
期望单板类型(Expected Board Type): G1FAN
固件版本(Firmware Version):
产品类型 (Product Name) : TNG1FAN (TNG1FAN01)
FPGA版本(FPGA Version):
序列号(Serial Number): 2102121961N0JC000081
BOM码(BOM Code): 2121961
软件版本(Software Version):
组件位置(Location): /ne=HUAWEI_8_238/sh=0/si=90
条形码(Bar Code):
硬件版本(Hardware Version): V
Shelf0-7-G2A212
Shelf0-100-TMF1PIU
Shelf0-11-G3TMD20
Shelf0-74-E3CTU
Shelf0-10-G3DAPXF
Shelf0-101-TMF1PIU
Shelf0-72-G3CXP
Shelf0-9-S7N502
Shelf0-103-G1EFI
HUAWEI_8_217
HUAWEI_8_217_subrack
out-of-service
BD_MODE_GENERAL
BD_MODE_LINE

Wrap Up

Team members:

Yanxia Tan(Chinaunicom)
Italo Busi(Huawei)
Xiao Li(Huawei)
Yanlei Zheng(Chinaunicom)
Aihua Guo(Futurewei)
Haomian Zheng(Huawei)
Henry Yu(Huawei)
Nigel Davis(Ciena)
Chaode Yu(Huawei)

First timers @ Hackathon:

Yanxia Tan
Xiao Li
Yanlei Zheng

- On our github we have: https://github.com/tanyanxia/IETF125-Hackathon-ivy-project
  - Full Video clip (7min)
  - JSON code
  - This presentation
- Plan in Vienna (Hackathon IETF126)
  - Continue implementing the model from other vendors (interests collected)